MIL-E-1/1643B(NAVY)
8 May 1975
SUPERSEDING
MIL-E-1/1643A(NAVY)
24 January 1969
(See note 16)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

1/ TYPE 8770

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR and shutter tube, frequency range 5,450 to 5,825 MHz, 250 kw

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb	Shutter circuit voltage	Pull in current	Ii	Pressurization	TA
Unit:	kw	Vdc	Vdc	mAdc	µAdc	psia	°C
Maximum:	250	-805	29.5	110	200	---	+65
Minimum:	---	-795	26.5	---	100	3	-20

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS: See note 3

Parameter:	Incident power	Du	prr	tpl	Ii	F
Unit:	kw	---	Hz	µs	µAdc	MHz
Test condition 1:	250	0.001	1,000	1.0	100	F2
Test condition 2:	8	0.001	1,000	1.0	100	F1

Frequency	
F1	5,450 MHz ±1%
F2	5,640 MHz ±1%
F3	5,825 MHz ±1%

GENERAL:

First article approval - Required

1/ Replaces type MCC-17A and BTR-546

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	First article inspection	2	-					
1011	Humidity	-	-		---	---	---	---
1016	Moisture resistance	-	-		---	---	---	---
1027	Temperature cycling	-	-		---	10	---	Cycles
---	Shutter holding current	5	-		Is	---	60	---
4021	Degradation due to vibration	4	-		---	---	---	---
1042	Shock, specified pulse	-	-	Test condition G	---	---	---	---
	Qualtiy conformance inspection, part 1	1	-					
1256	Electrode current (shutter)	-	-	Es = 28 Vdc	Is	---	156	mAdc
---	Pull in current	-	-		---	---	110	mAdc
4401	Ignitor ignition time	-	-	Ebb = -800 Vdc	t	---	5	sec
4416	Insertion loss (1)	-	-	F = F2 ±0.1% (shutter open)	Li	---	0.7	dB
4416	Insertion loss (2)	-	-	F = F2 ±0.1% (shutter closed)	Li	26	---	dB
4473	Low level VSWR	-	-	F = F1, F2, F3 ±0.1%	σ	---	1.4	V
4452	Flat-leakage power (1)	15	1		pf	---	70	mW
4452	Flat-leakage power (2)	15	2		pf	---	70	mW
4452	Spike-leakage energy (1)	15	1		Ws	---	0.25	ergs
4452	Spike-leakage energy (2)	15	2		Ws	---	0.25	ergs
4471	Recovery time(1)	8	1	Time for Li to reach 3.8 dB	t	---	12	µs
		8	1	Time for Li to reach 0.8 dB	t	---	19	µs
4260	Noise figure	10	-	F = F2; Li = 100 µAdc	ΔNF	---	0.2	dB
4278	Phase shift	12	-		---	---	±3	Degrees
	Quality conformance inspection, part 2							
4471	Recovery time (2)	9	2	Time for Li to reach 3.8 dB	t	---	5	µs
		9	2	Time for Li to reach 0.8 dB	t	---	12	µs
---	Breakdown power	11	-	F=F2; Li = 100 µAdc	---	---	200	mW
---	Ignitor oscillation	7	-		Ii	---	60	µAdc
---	Shutter cycling	6	-	F=F2; Es = 28 Vdc	---	250	---	Cycles
	Quality conformance inspection, part 3							
---	Life test	13	1	Group D	t	1,000	---	hrs

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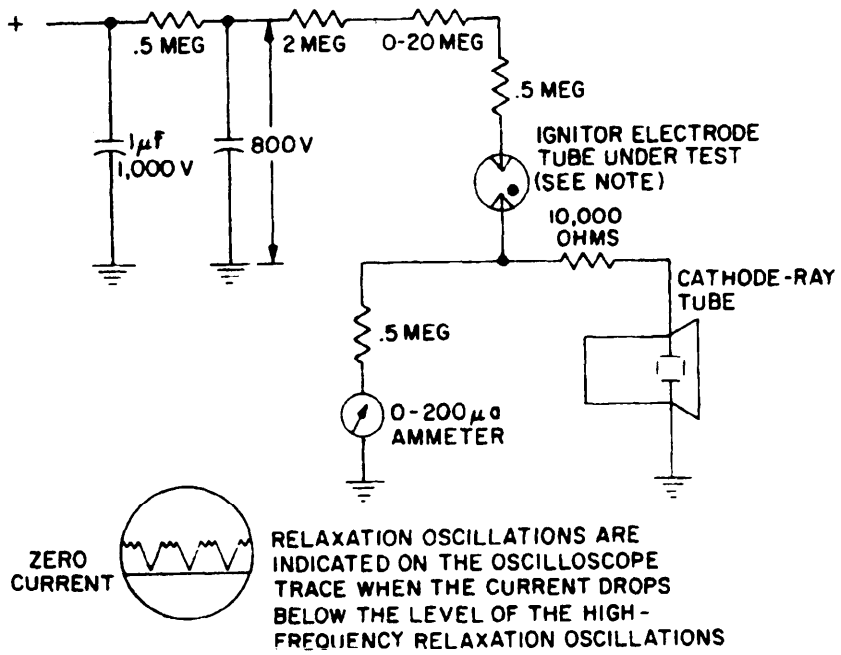
METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
---	Life-test end points:							
4416	Insertion loss (1)	-	-	F = F2 ±0.1% (shutter open)	Li	---	1	dB
4416	Insertion loss (2)	-	-	F = F2 ±0.1% (shutter closed)	Li	25	---	dB
4278	Phase shift	12	-		---	---	±5	Degrees
4471	Recovery time (1)	8	1	Time for Li to reach 3.8 dB	t	---	14	µs
		8	1	Time for Li to reach 0.8 dB	t	---	21	µs
4471	Recovery time (2)	9	2	Time for Li to reach 3.8 dB	t	---	8	µs
		9	2	Time for Li to reach 0.8 dB	t	---	15	µs
---	Crystal degradation	14	-		NF	---	2	dB
---	Shutter cycling	6	-	F = F2 ±0.1%; Es = 28 Vdc	---	50,000	---	Cycles

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. First article inspection shall consist of performing all tests listed on this tube specification sheet. The sample size and allowable defects shall be in accordance with MIL-E-1, Qualification, Samples, and acceptance criteria. Three copies of the test report, prepared in accordance with MIL-STD-831, shall be forwarded to the pruchasing activity for evaluation by the preparing activity.
3. All low and high level tests shall be made with the tube shutters open unless otherwise specified.
4. The shutters shall be held open by the minimum holding current specified herein during vibrations. The tube shall be vibrated along a line parallel to the axes of the shutters. Insertion loss test setup shall be used with the unit under test mounted on the vibrator in such a manner that a standard waveguide section of similar geometry may be substituted for the shutter tube. The modulation, due to vibration, shall not be greater with the shutter tube in the line than with the waveguide section. Tubes shall meet all applicable requirements of this specification sheet after vibration.
5. The current required to hold the shutters open shall be within the limit specified herein when the tube is subjected to a hammer blow, 15 G, directed along a line parallel to the axes of the shutters and against that side of the tube which is directly opposite the side on which the shutter coils are mounted.
6. Each time the shutters open, a low-level signal goes through the tube and is registered by a counting circuit. At the end of this test, the number of signals registered shall equal the number of pulses applied to the shutter coils.
7. The tube shall be placed in the circuit shown, or equivalent circuit. With the supply voltage adjusted to 800 Vdc, the ignitor current shall be set by means of the variable resistor in series with the ignitor electrode. The minimum current that will prevent low-frequency relaxation oscillations shall be within the limits specified on the TSS. The high-frequency oscillations cause no difficulties. Relaxation oscillations are indicated on the oscilloscope trace when the current drops below the level of the high-frequency relaxation oscillations.

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NOTES: -Continued



8. After the trailing edge of signals, the tube shall "deionize" to an insertion loss of 3.8 dB within 12 μs and to an insertion loss of 0.8 dB within 19 μs.

9. After the trailing edge of signals, the tube shall "deionize" to an insertion loss of 3.8 dB within 5 μs and to an insertion loss of 0.8 dB within 12 μs.

10. The change in low-level noise figure of the tube shall be less than 0.2 dB with the ignitor "off" to "on". The noise figure is defined as the signal-to-noise-ratio in over the signal-to-noise-ratio out.

11. Breakdown power is defined as the maximum power level transmitted through the tube before a microwave gas discharge is initiated within the tube such that the power transmitted through the tube is sharply attenuated.

12. With the specified ignitor current flowing, the total electrical length of the single TR tube shall be measured every 100 MHz over the frequency range 5,450 to 5,825 MHz. The electrical length of each TR tube shall not deviate by more than ±3 degrees from the reference as shown in the curve of the graph, figure 2. Phase shift shall be defined as:

$$= 188.0 + \left[0.400 + 0.013 \sin \frac{\pi(F1)}{375}\right] \left[F1\right]$$

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13. At the end-life point of the tube, all other requirements of this specification sheet shall be met. Two crystal failures will be permitted during the 1,000-hour test.

14. Crystal degradation. When the tube is terminated in a crystal mixer containing a 1N23WE crystal, the end-life point of the tube shall occur when the crystal noise figure has degraded by more than 2 dB. The crystal degradation shall be based on noise figure measurements initially not greater than 7.5 dB.

15. Direct measurement of flat-and spike-leakage energy shall be made as follows:

The leakage power shall be measured directly from a visual display on a fast writing oscilloscope (50 MHz bandwidth) obtained by applying the output of a crystal detector to the vertical deflection system of the oscilloscope. The vertical deflection system and crystal output are first calibrated against known power levels. This shall be done with precision variable attenuators, thermistor, and power meter. A match shall be maintained between the detector output impedance and the oscilloscope input impedance. The peak flat-leakage power can then be determined by noting the maximum amplitude of the flat portion of the leakage pulse and compared to the calibration data. Spike-leakage energy is determined by measuring the peak spike amplitude and the spike width at the 3 dB (half power point). The spike-leakage energy is then computed as follows:

$$WS = ts_{3dB} \times Ps \times 10^{7}$$

Where:
Ws = spike leakage energy in ergs/pulse.
ts 3 dB = spike width at half the spike amplitude in seconds.
Ps = maximum spike amplitude in watts.

16. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N045)

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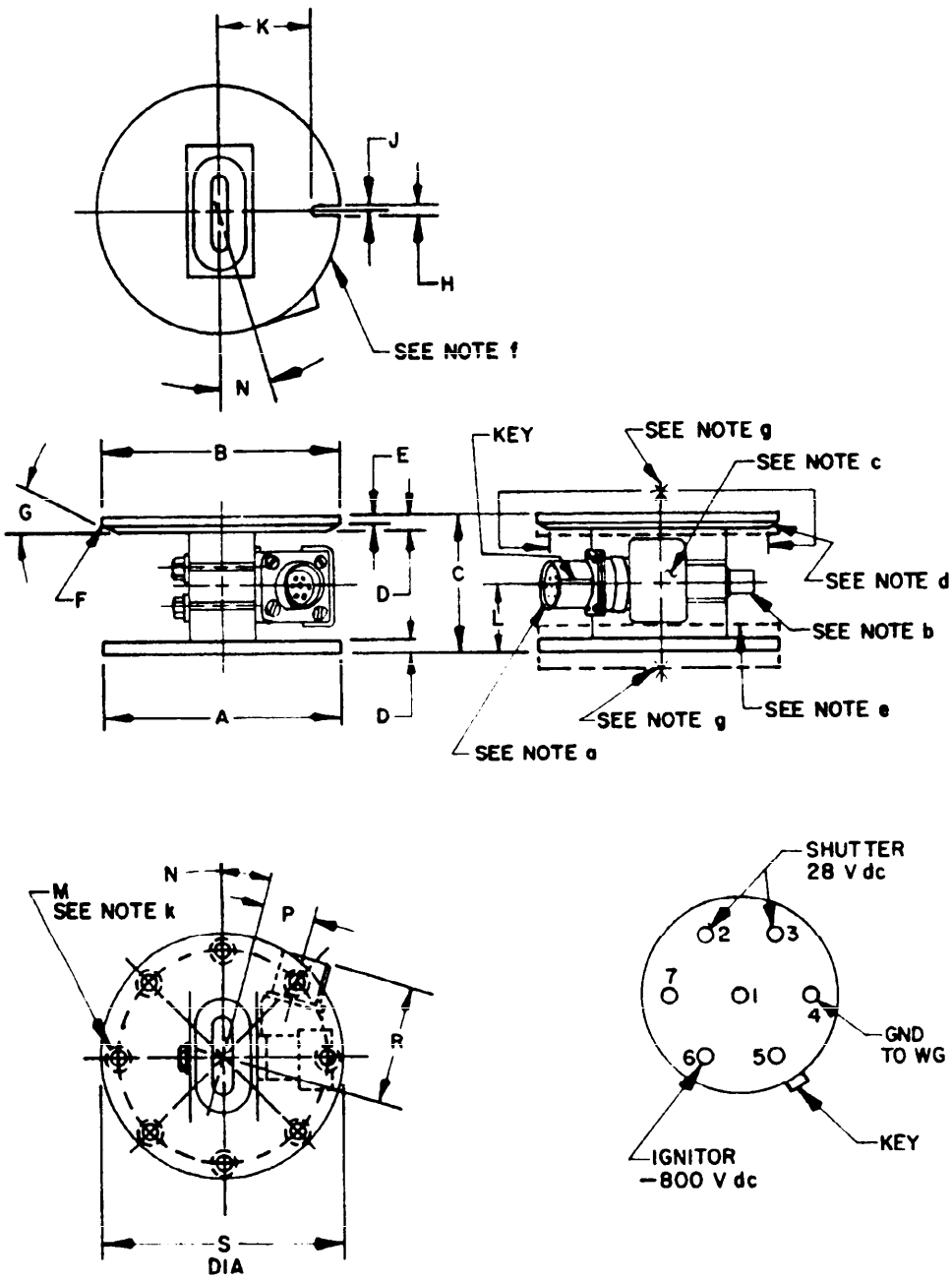


FIGURE 1. Outline drawing of electron tube type 8770.

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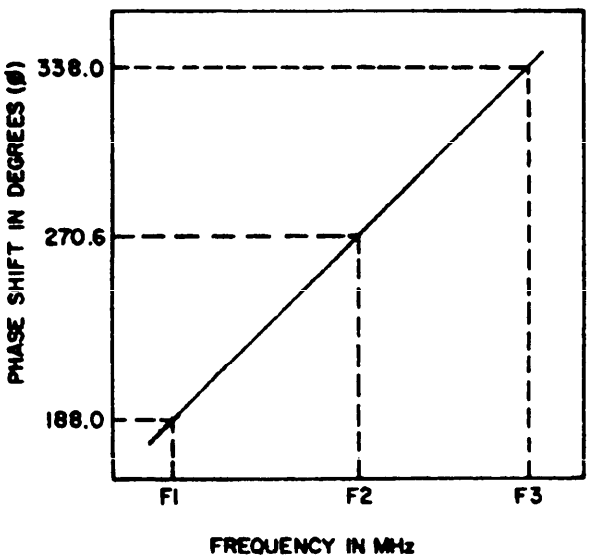
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (See note 1)		
C	2.100 (53.34)	2.120 (53.85)
K	1.527	1.537
L	1.0475	1.0775
M	.10-32 NC-2B	
N	14°	16°
P	.7825	.9075
R	1.6250	1.750
S	3.248 (82.49)	3.252 (82.60)
Quality conformance inspection, part 2		
A	3.610	3.640
B	3.595	3.625
D	.235	.265
E	.135	.145
F	.0313	.0625
G	19°	21°
H	.130	.137
J	.063	.067

NOTES:
a. Deutsch connector DM 9606-7PB, or equivalent.
b. Exhaust stem shall not extend beyond O.D. of flange.
c. Ignitor resistor encapsulated in potting material.
d. Input transmitter flange-maintain .188 clearance behind flange for attaching clamp.
e. Output receiver flange-maintain .156 clearance behind flange for hardware protrusion.
f. Flanges - steel, nickel-plated.
g. Omit paint finish.
h. Flanges shall be parallel within 0.003 TIR.
k. Drill thru and tap 8 holes equally spaced.

FIGURE 1. Outline drawing of electron tube type 8770 - Continued.

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$$\phi = 188.0 + \left[0.400 + .013 \sin \frac{\pi (F1)}{375}\right] \left[F1\right]$$



2. 110 inches blank reference:
F1 = 286.0 degrees
F2 = 301.6 degrees
F3 = 315.0 degrees

FIGURE 2. Phase shift curve.

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